

TyreMind — what to say, and when

Seven minutes, ten beats, one screen. Open the app on **The pitch** and scroll as you talk. Do not switch screens. Everything you need is on this one page.

Words in **bold** are the lines worth saying close to verbatim. Everything else is yours to paraphrase. Lines in *italics* are stage directions, not speech.

Before you start

Have the app open at #/pitch, race set to Monza. Scrolled to the top.

Say nothing about the technology in the first thirty seconds. Say what goes wrong without it.

00 · The hook — 30 seconds

Top of the page. Do not scroll yet.

A Formula 1 race is decided by one number that nobody can measure.

How fast the tyre is going away. Get it right, you win the pit stop. Get it wrong, you lose the race in the pit lane, and you lose it to a decision you made three laps too early.

Around ten teams in the world can measure it properly. **Everybody else is guessing.**

Pause. Then scroll to beat 1.

01 · Why it is hard — 45 seconds

The stacked chart. Point at the orange band.

Here is why it is hard. **Four things move a lap time at the same time.**

The orange is the tyre wearing out, slowing the car down. It grows every lap.

The blue underneath is the fuel burning off. The car is getting lighter, so it is getting faster — by almost exactly the same amount, in the opposite direction.

Point at the dashed line.

And that is what the timing screen shows you. The two cancel out. The one number everybody watches barely moves, while underneath it the tyre is dying.

02 · The physics check — 45 seconds

The car diagram. This is your credibility beat. Slow down here.

Before I show you what we built, let me show you how you know the physics underneath is real.

These are the four tyres on the car, seen from above. The darker ones did more work. **Lean on right-hand corners all lap and the left-hand tyres carry the load.**

Point at the green box.

So from tyre wear alone, our model says this circuit runs clockwise. It does run clockwise. **We never told it. It works that out on seven of the eight circuits we tested.**

If the physics underneath were wrong, it could not do that.

03 · What we built — 40 seconds

The narrowing band.

Now the product. We separate the four causes and report only the tyre.

Trace the band left to right with your finger.

Watch the grey band close. On the left the model has seen almost nothing, so it is honest about being unsure. As the laps come in, it narrows.

That is a model learning from evidence rather than asserting a number. **It is the difference between a guess and a decision.**

04 · The forecast — 30 seconds

The fan.

Then it tells you where the car will be in three laps, and in fifteen.

Notice the band gets wider the further out it looks. It should. Anyone who draws you a confident line twenty laps into the future is lying to you, and a strategist finds that out the hard way.

05 · The decision — 40 seconds

The pit window curve. This is what they buy.

And this is the thing a team actually pays for.

Every lap he could stop on. For each one we simulate the rest of the race over and over and count how often that lap turns out to have been right.

Point at the dip, then the shaded block.

The lowest point is the answer. The shaded block is the window a strategist keeps open. Not a number — a window. That is how the decision is actually made on a pit wall.

06 · Why a simpler tool cannot do this — 30 seconds

The four bars.

You might ask why this needs anything clever. Here is why.

We measured the shape of nearly three thousand real stints. **Only just over half fade evenly.** The rest warm up, recover, or fall off a cliff — and the cliff is the one that ends races.

Every competing model assumes a straight line. So it is the wrong shape for almost half of real racing. Ours was never told what shape to expect.

07 · Trust, part one — 30 seconds

The consensus rows.

Two reasons you can trust the number.

First: we do not ask the model once. We re-run it under deliberately different assumptions about the things nobody can measure.

Point at the overlap.

When four disagreeing starting points land on the same answer, that answer belongs to the race, not to our choices. When they scatter, we say so instead of picking the one we like.

08 · Trust, part two — 50 seconds

The calibration diagonal. This is your strongest beat. Do not rush it.

Second, and this is the one I would want to hear if I were you.

The dashed diagonal is a tool being exactly as confident as it deserves to be.

Point at the orange lines sagging below it.

Every competing model sits below that line. They all claim more certainty than they have earned. That is the failure that gets a strategist hurt.

Ours did too. It told us it was right ninety-five times in a hundred. We checked it against sixty-nine thousand real laps. It was right seventy-five.

So we rebuilt the confidence from measured outcomes instead of theory. *Point at the green pair on the diagonal.*

That is us now.

Almost nobody runs this test. That is why almost everything on the market overstates how sure it is.

09 · Bigger than racing — 30 seconds

The scatter.

One more thing. Strip the motorsport words away and this is a general problem. **Something wears out while it works, you cannot measure it directly, and other things move the only signal you can see.**

We pointed the identical code at NASA's jet engine benchmark without changing a line. Each dot is an engine. **It is not a Formula 1 trick.**

10 · The business — 60 seconds

The four pillars, then the closing line.

So who buys it.

Roughly ten teams in the world can do this today — they have departments, licensed data and budgets. **Every series below them makes the same pit calls, on the same compounds, from the same public timing feed, with nothing.**

F2, F3, F4, regional single-seaters, GT, endurance privateers, club racing.

The capability is not new. It has simply never been cheap enough to reach them.

Point at the pillars.

It runs on a laptop a team already owns. No GPU, no cloud bill, and the data it needs is public. It needs four fields — lap time, tyre type, tyre age, lap number — and **every series already publishes all four.**

And nothing leaves the garage. No upload, no vendor sees their data. In motorsport that is not a feature. It is the condition of sale.

The close — 20 seconds

Stop scrolling. Look up.

Everyone can build a dashboard that shows a tyre wear number.

We built one that is right, that tells you how confident it is and is genuinely right that often, and that we then tried very hard to break ourselves.

Thirteen of our own ideas failed. We published all thirteen. Including a far more advanced physics model we built specifically to try to beat our own system. It lost, so we did not ship it.

We would rather you checked us than believed us.

If they interrupt

Short answers. Do not over-explain. Offer the screen.

"Is this live or a recording?"

Live. *Change the race in the sidebar.* Every chart on the page just refitted.

"How do you know you are right, if nobody publishes real tyre wear?"

We do not need the right answer to prove a method is wrong. A method that says tyres get faster as they wear has disproved itself, and that happens in about three races out of four. That result needs no trust at all.

"What is your accuracy?"

Four point three times better than the closest published research model at recovering the wear rate. But the number I would rather you asked about is the confidence one — it is on screen, and we are the only ones who measured it.

"Where are you weakest?"

A simpler model forecasts raw lap times better than we do, and we send that question to it rather than pretending otherwise. On real pit stops we tie with the closest published research rather than beating it. Both are on the evidence screen.

"Who is your competition?"

Inside Formula 1, the teams' own departments, and we are not trying to displace those. Below Formula 1 there is nothing, which is the whole business.

"How long did this take / is it finished?"

It runs offline from a clean clone today. Six hundred and fifteen automated tests, thirty-two recorded experiments, every result file in the repository.

"What would you do with investment?"

Two things. Get it into one junior championship as a pilot, and build the hierarchical layer that lets a compound parameter carry across a season properly — we tested one version of that and it failed, and the result file says so.

Things not to say

- Do not say **Kalman filter**, **state-space**, **conformal** or **posterior** unless someone asks a technical question first. They are correct and they end a non-technical listener's attention.
 - Do not claim to beat a professional strategist. The measured margin is small and the honest claim is against the standard method.
 - Do not oversell the market. You have no revenue figures and inventing one is the fastest way to lose a room that has been checking everything else.
 - Do not skip the failures. Thirteen refuted ideas is your credibility, not your weakness.
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Timing

Beat	Target
Hook	0:30
01 Why it is hard	0:45
02 Physics check	0:45
03 What we built	0:40
04 The forecast	0:30
05 The decision	0:40
06 Why simpler fails	0:30
07 Trust, consensus	0:30
08 Trust, calibration	0:50
09 Bigger than racing	0:30
10 The business	1:00
Close	0:20
Total	7:30

If you are cut to five minutes, drop beats 04, 06 and 09. Never drop 02, 08 or 10.